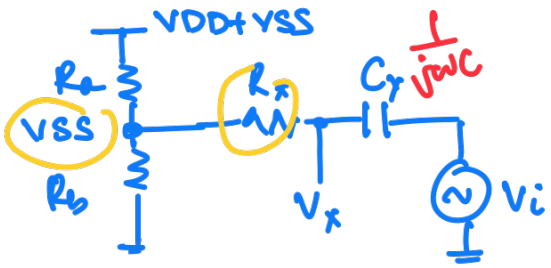
Lecture #09



$$V_x = \frac{\frac{1}{sC_y}}{R_x + \frac{1}{sC_y}} VSS + \frac{R_x}{R_x + \frac{1}{sC_y}} V_i$$

$$= \frac{1}{1+sC_yR_x} VSS + \frac{sC_yR_x}{1+sC_yR_x} V_i$$

$s=0$

I want it to be 1

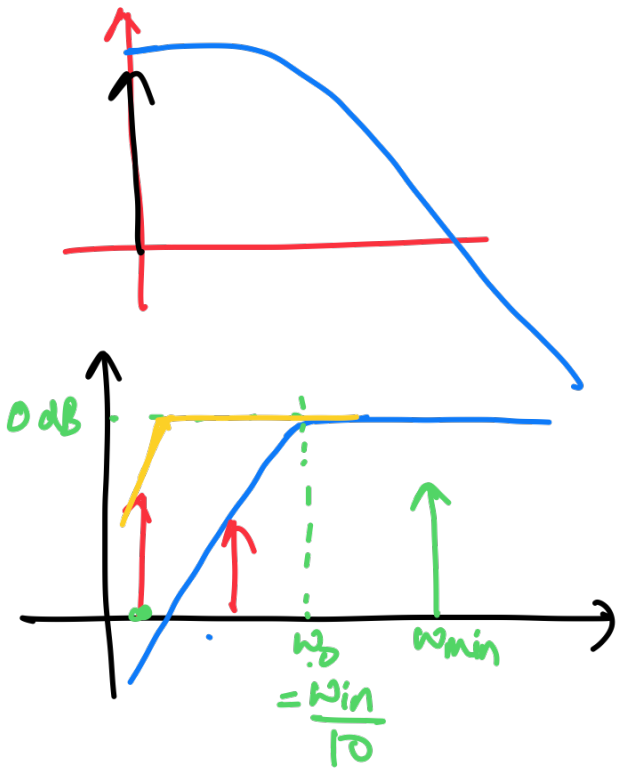
$$\left|\frac{sC_yR_x}{1+sC_yR_x}\right| \neq 1 \Rightarrow \left|\frac{j\cdot 10}{1+j10}\right| = \frac{10}{\sqrt{101}}$$

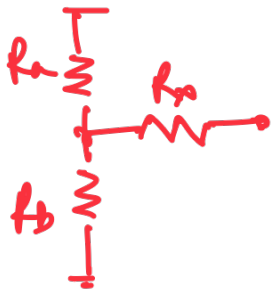
$$\Rightarrow |sC_yR_x| \gg 1$$

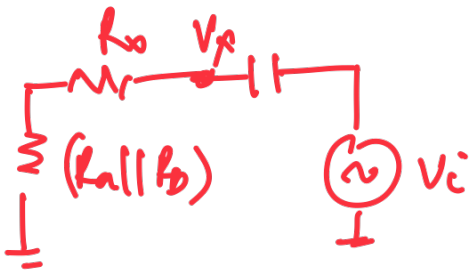
$$|j\omega C_yR_x| = 10$$

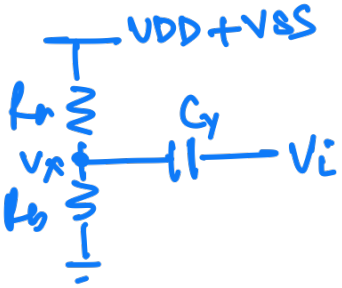
$$C_y = \frac{10}{\omega R_x}$$





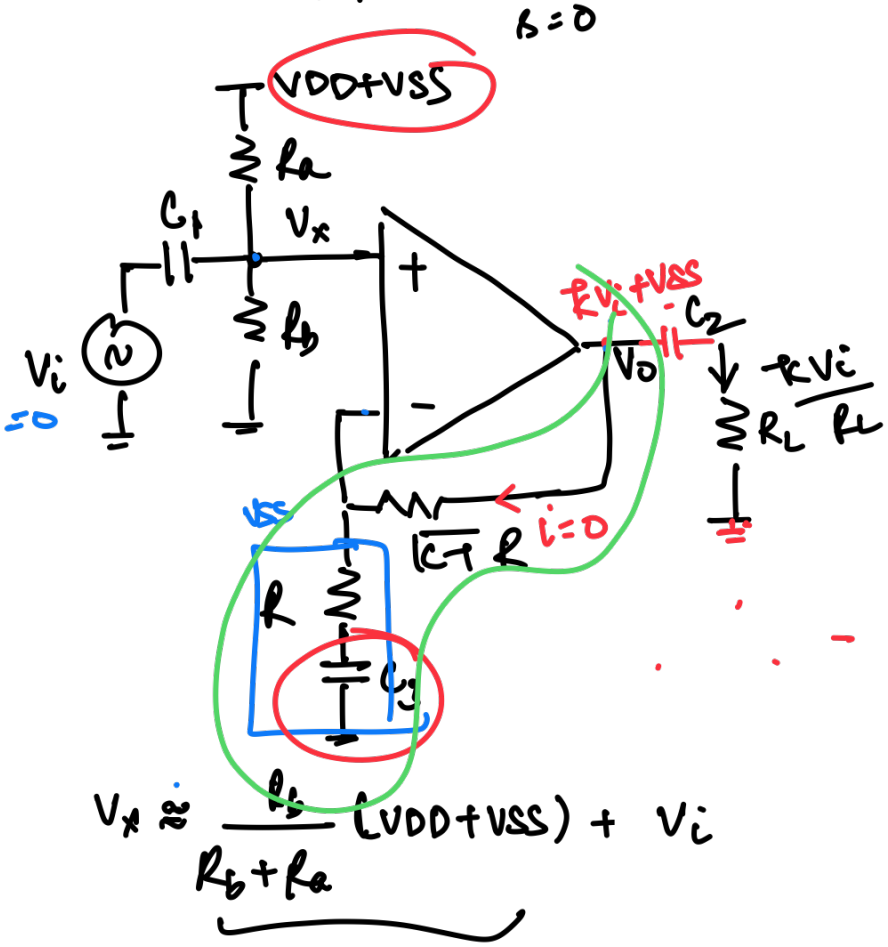






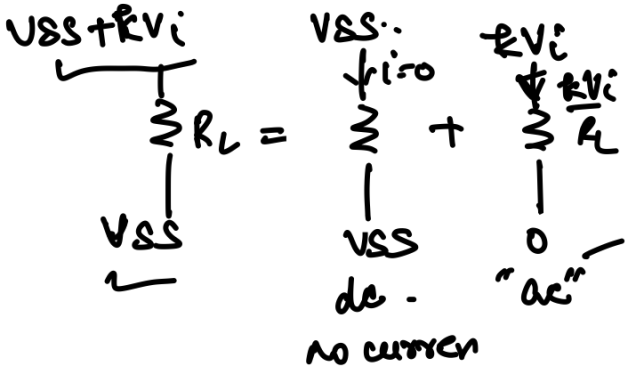
$$V_x = \frac{(\frac{1}{sC_y} \| R_b)}{R_a + (\frac{1}{sC_y} \| R_b)} \cdot (VDD+VSS) + \frac{(R_a \| R_b)}{\frac{1}{sC_y} + (R_a \| R_b)} V_i$$

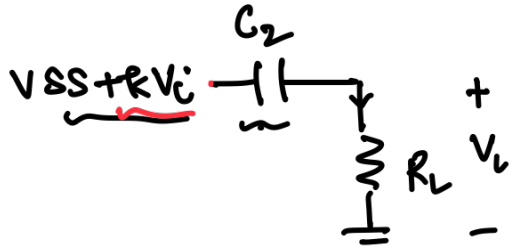
$$\left.\frac{\left(\frac{1}{sC_1}\,\|\,R_b\right)}{R_a+\left(\frac{1}{sC_1}\,\|\,R_b\right)}\right|_{s=0} = \frac{R_b}{R_b+R_a}$$

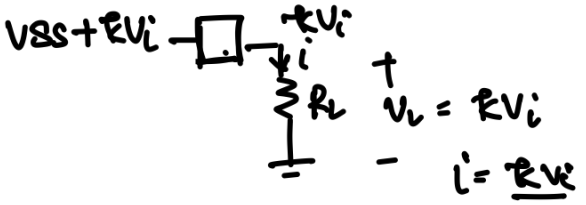


$$V_x \approx \frac{R_b}{R_b+R_a}(VDD+VSS) + V_i$$

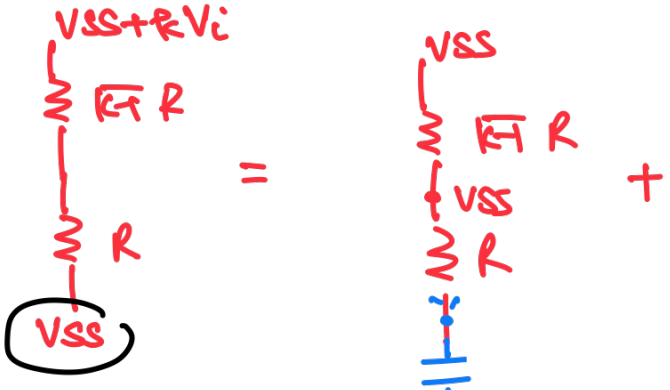
$$= VSS + V_i$$

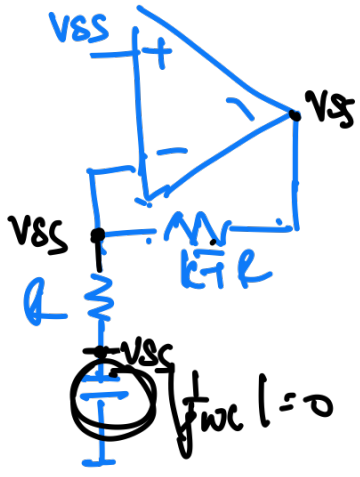


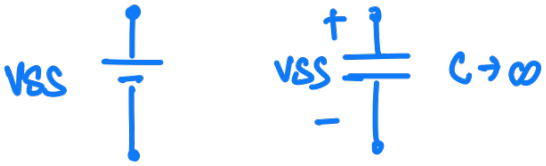


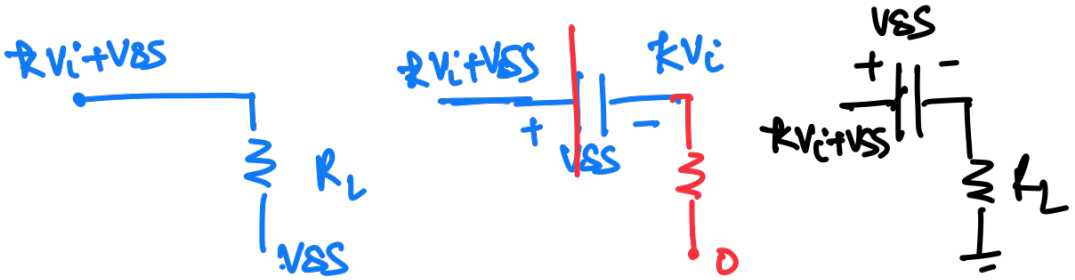


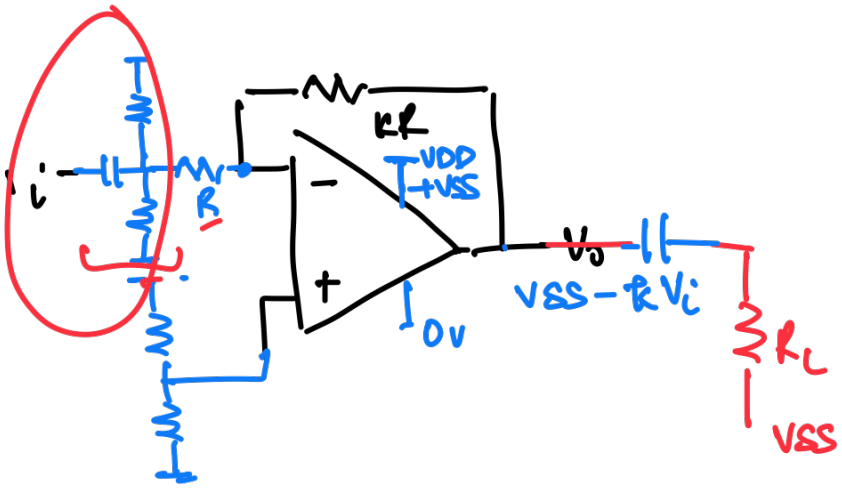
$$\left|\frac{V_o}{kV_i}\right| = \left|\frac{sR_1C_2}{1+sR_1C_2}\right| \simeq 1$$

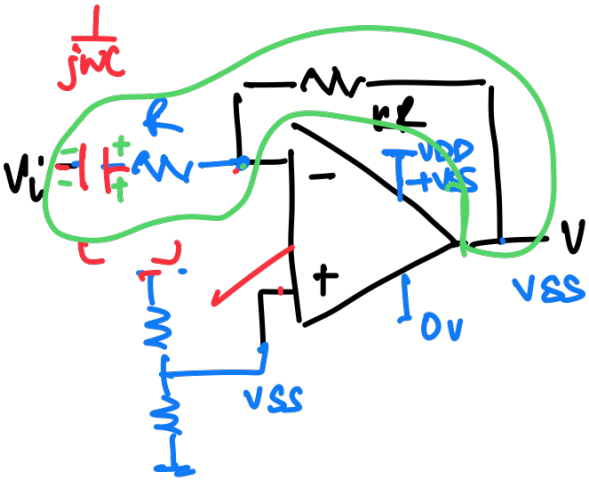


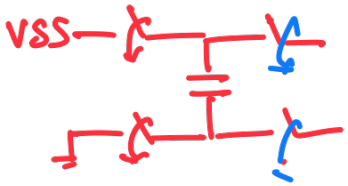




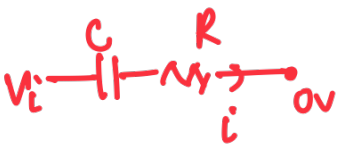




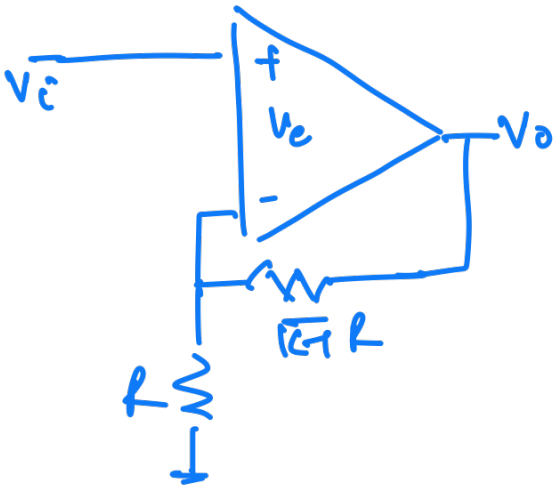




"Switched-capacitor circuits"



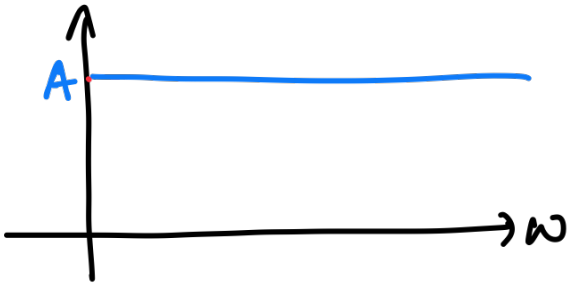
$$i = \frac{V_i}{R + \frac{1}{sC}} = \frac{V_i}{R} \cdot \frac{sCR}{1+sCR}$$



$A \to \infty \Rightarrow V_e = 0 \Rightarrow V_o = \beta V_i$

$$\frac{V_o}{V_i} = \frac{k}{(1+\beta k)}$$

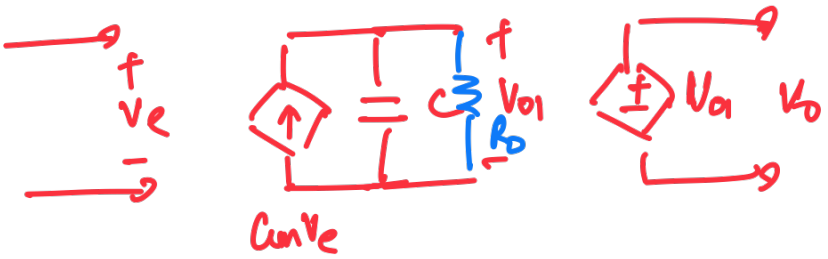
$V_e \to 0$



V_i V_o

$$\frac{V_o}{V_i} = \frac{-1}{sCR}$$





$$\frac{V_o}{V_e} = \frac{G_m}{sC} \rightarrow \frac{G_m R_o}{1+sCR_o}$$